

Feature Engineering and Preprocessing on the Titanic Dataset

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Dataset Chosen

Dataset: Titanic Survival Dataset (Kaggle)

Type: Classification

Goal: Predict whether a passenger survived or not.

Feature Engineering Techniques Applied

- Missing Value Handling: Median for Age, Mode for Embarked
- Encoding: Label Encoding (Sex), One-Hot Encoding (Embarked)
- Scaling: StandardScaler for numeric features
- Feature Extraction: PCA with 2 components
- Feature Selection: SelectKBest with ANOVA F-test

Observations

- Missing data handled; all features converted to numeric form.
- Top features: Sex, Fare, Pclass, Age, SibSp.
- Scaling and PCA improved dataset readiness for ML models.
- PCA retained approximately 75% variance while reducing dimensionality.

Impact of Transformations

Before preprocessing, the data had missing values and inconsistent scales. After feature engineering, it became clean, numerical, and

standardized — ready for model training. Feature selection and PCA helped remove noise and improved interpretability.

Ethical Discussion

Including sensitive features like gender or marital status in predictive models can introduce bias. To mitigate such bias:

- Avoid using sensitive attributes as predictors when possible.
- Test for fairness metrics (e.g., disparate impact ratio).
- Apply debiasing methods such as reweighing or fairness-aware learning.
- Use interpretability tools like SHAP or LIME to ensure transparency.

Summary

This project demonstrates end-to-end data preprocessing — handling missing values, encoding, scaling, and dimensionality reduction. The final dataset is clean, unbiased, and ready for modeling.